

Preface

You’ve almost certainly already asked ChatGPT or another artificial intelligence (AI) tool a coding question and saw the tool spit out (seemingly) perfect code within seconds.

This book explores the potential and limitations of AI tools in software development. Let’s start with the positives. We’ll show you that ChatGPT, GitHub Copilot, and others can do more than just generate a few lines of code. You can also use them to create unit tests, search for errors, perform refactoring tasks, develop database schemas, optimize SQL queries, administer servers, and write scripts.

But the brave new AI world also has its downsides. Even if the code looks plausible and the accompanying text is elegantly formulated, some AI suggestions are simply wrong due to logic errors, the use of nonexistent functions, or references to variables whose names are actually different. Although the code does work sometimes, it’s inefficient or insecure.

AI tools make mistakes—that much can’t be disputed. Used correctly, however, these new tools can also save you a lot of time. That’s what this book is all about.

Local Execution of Language Models

Companies in particular often have problems entrusting their code base to tools that run in the cloud. Even if AI providers promise not to use your code for the training of future language models in company offers, the question arises as to how far you can trust these promises and whether data protection problems may arise.

The easiest way out is to run the language models required for AI tools locally. You can’t download the large language model (LLM) from ChatGPT, but various alternatives are available free of charge that work similarly well.

However, the local execution of LLMs consumes a lot of resources, so only powerful notebook computers or PCs are suitable for this. New CPUs with neural processing units (NPUs) promise new possibilities in this respect. However, a real benefit for the local execution of LLMs only arises in combination with suitable software—and there is still a lot to be done.

There is one more option: you can set up your own server in your company or in a data center that is available to all employees for AI tasks. In that case, “local” doesn’t refer to your own notebook computer, but to a company-owned computer. In this book, we’ll show you how you can use local language models and even optimize them with your own training material.

We’ll also present the *GPT4All*, *Ollama*, and *Tabby* programs to run free LLMs locally. You can then use the local language models in chat mode for coding (e.g., by using the *Continue* editor plug-in) or via an application programming interface (API) to develop your own AI applications.

Coding with AI Support for Advanced Users

If you’ve asked ChatGPT for advice on a coding problem or if you’ve followed suggestions from GitHub Copilot (or a similar tool), then you know about interactive AI support for coding. But there’s more to it! You can go one step further by using the appropriate libraries:

- **Processing many code files automatically**

You want to change all comments in 100 files from German to English? You want to migrate

a project from Python 2 to Python 3? Instead of interactively processing file by file, couldn't you write a script that automates this task? We've tried it out and will share our experiences with you.

- **Level 3 tools**

Similar to self-driving cars, there are also categories of autonomy for AI coding tools. ChatGPT, Claude, GitHub Copilot, and other tools are assigned to levels 1 and 2 depending on their application (*code completion* or *code creation*). However, the first level 3 tools (*supervised automation*) already exist that go beyond this and independently generate or modify code files according to your instructions. We experimented with *OpenHands* and *Aider*, and we were particularly enthusiastic about the *Aider* concept.

- **Retrieval-augmented generation (RAG)**

It's currently impossible for private users or small businesses to train their own language models. By using RAG, however, you can add your own data to language models. This results in a host of additional application options, including the integration of daily updated data or the automated conversion of text prompts into SQL queries. In this chapter, we focus primarily on the *LlamaIndex* library.

Theoretically, the tools outlined here represent the next development step in the use of AI in coding. In real life, however, our experiences have been mixed. Not everything worked out as we had hoped. AI is already able to perform some of these tasks, but for more complicated requests, patience is still required until these tools become suitable for everyday use.

AI Makes Mistakes and Hallucinates

If you talk to other developers, you'll hear many anecdotes about the major mistakes AI tools have made recently. There are also many chapters in this book in which we show examples with faulty AI code, incorrect conclusions, or simply invented functions or options. (Invented functions have to do with an unpleasant feature of current language models: If AI can't give a reliable answer due to insufficient training material, it invents a text that sounds as plausible as possible—that is, AI “hallucinates.”)

Depending on how you use AI tools, the error rate varies between rather low (widespread programming languages, everyday problems) and quite high (exotic languages, new libraries/APIs, very specific tasks). The biggest problem isn't that mistakes happen. We all know that people also make mistakes, probably even more often. What is annoying is that AI tools argue even the biggest nonsense politely and at a pretty high linguistic level. That's what makes it so difficult to tell the right answers from the wrong ones. In our view, AI tools will only be truly “intelligent” when they answer: “I don't know” or “I'm not sure about this proposed solution, but it could work like this.” We aren't that far yet in early 2025, even if there are efforts to show how the answers can be derived.

As you can see, we aren't trying to sell you an AI tool! On the contrary, we point out the issues we've encountered in all chapters, especially in [Chapter 13](#).

The Three Stages of Using AI

While writing this book, we not only worked intensively with AI tools ourselves but also spoke to many other developers. Through this, we've recognized a pattern in the use of AI tools in coding. It corresponds fairly closely to the *hype cycle*, a model coined by consultant Jackie Fenn to describe how new technologies are perceived and used (see [Figure 1](#)):

- **Euphoria**

The first experiments will leave you, like most developers, speechless with enthusiasm. This “thing” understands your questions, gives answers that make sense, and even explains the code. That's crazy!

- **Disillusionment**

Then, the moment arrives when the AI tool of your choice makes its first mistakes. It's no

longer about Hello World functions, but about real, complex code. The AI tool explains the world to you with the utmost confidence, and you realize that the explanations are convincing in terms of language, but completely wrong in terms of content. You might conclude here that AI tools are useless, their use irresponsible, and so on.

- **Pragmatic use**

After a few weeks, you'll develop a good feeling for when AI results are most likely correct and where you need to make improvements. You're getting better and better at assessing the limits. What remains is a tool that makes mistakes, but is still a real help within the scope of its capabilities. You question every AI answer and reckon with an error rate of perhaps 10%. But you use what works—and that's quite a lot.

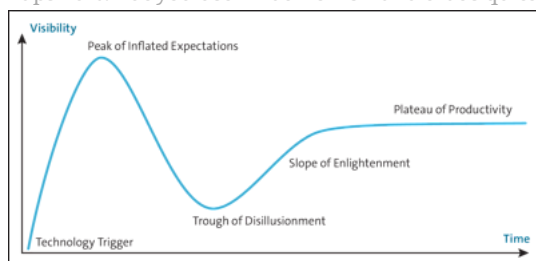


Figure 1 AI Tools Currently in the Stages of the Hype Cycle

There's No Getting around AI Tools

Despite all the problems, we're convinced that AI tools are here to stay. According to one study (<https://www.atlassian.com/software/compass/resources/state-of-developer-2024>), managers believe that the time-saving factor is huge, while developers who actually use AI tools are much more cautious (and realistic) about the benefits!

The extent of the efficiency gain depends heavily on the type of code you develop. But there is always a certain amount of time saved and a higher level of convenience at work. Regardless of whether you see the glass as half full or half empty, coding using AI support increases your efficiency today and will increase it even more tomorrow.

Chemist Derek Lowe put it this way in February 2023:

I don't think AI and automation will replace chemists, but rather that the human chemists who use them well will replace the ones who don't.

We're convinced that this statement, which has often been quoted since, also applies to the IT world. AI therefore won't replace software developers, at least not in the near future. But developers who use AI tools in a reasonable way will replace their colleagues who do not.

Now is the right time to get to grips with AI tools! Yes, AI tools will become even better in the coming years, but the potential is already huge. And the knowledge and experience you gain today will help you better understand new AI technologies tomorrow and integrate them more quickly into your day-to-day work.

According to the 2024 Developer Survey (<https://survey.stackoverflow.co/2024/ai>), 63% of all developers already use AI tools. An additional 13% are planning to use them in the near future. We're clearly at the beginning of a new era in terms of software development. Despite all the fears associated with such far-reaching changes, we've found the use of AI tools in our day-to-day coding work very satisfying. AI tools haven't done the thinking for us, but they have done a lot of boring typing. The concepts of the software we developed still came from us, but the often-tedious work of implementation has become easier for us.

With this in mind, we hope you enjoy reading this book. Be inspired by our examples, and look for ways to integrate AI tools into your day-to-day work in a meaningful way. Take advantage of the opportunities that arise here!

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PS: This book deals exclusively with the use of AI tools for software development and administration. Clearly, the new developments in the AI sector will have a wide-ranging impact on society, both positive and negative. However, this isn't a topic in this book, nor is the use of AI tools outside of IT.

Sample Files

There are sample files for a few chapters of this book, which you can download here:

www.rheinwerk-computing.com/6058.

